

Automated Feature Extraction RFI Objective Questions

Objective 1: Develop and Deploy a High-Performing Automated Feature Extraction (AFE) Capability:

What AFE automation processes do you utilize to produce features today? What sources of information are required for your change detection solution? For a given source (or combination of sources), what is your methodology to create real world representations of specific features? What combinations of data types are used in your process (e.g. vector-to-vector, image-to-vector, PDF-to-vector, and/or image-to-image, image-to-surface model, etc.) for specific features? How do your current processes ensure accurate position, confidence assessment and time is associated with automated feature extraction processing? Can you describe how your AFE solution(s) fits into geospatial data production tools and systems utilizing multiple input data formats? How do your capabilities integrate with other commercially available geospatial tools? What Technical Readiness Level¹ (TRL) is your capability current graded at? Describe the process on how you would separate out multiple features that may be conflated during AFE processing.

Objective 2: Ensure High-Quality Data and Enable Analyst-in-the-Loop Validation:

How would you ensure that the changes that may occur during the feature extraction processing are detected and retained are relevant to geospatial data? What source formats would your system or algorithm(s) ingest to produce automated specific and/or multiple features within the same area of interest? What processes would you employ to provide GEOINT assurance that changes are verified, comprehensive and valid? Describe your process to organize an analyst notification system and how would you ensure critical processes/outputs are prioritized for review? What are your defined threshold metrics for minor, major and critical for AFE outputs for topographic, maritime and aeronautical features? How would you achieve an objective threshold accuracy of 80% with a goal for objective accuracy of 99% for topographic, maritime and aeronautical features? How does your solution interface with GIS systems for viewing, evaluating and editing results? Describe any AI/ML models that are used for your quality assessments.

Objective 3: Implement Automated Change Detection to Maintain Data Currency:

Does your AFE capability describe the nature or type of change detected to include comparing feature(s) to new source data against existing feature databases? Please discuss any plans for automated assessment of change detection recommendations as applied to a multi-mission community. Do you have any capabilities that manage change at the data level that could be used, for example, to determine if an identified change is currently represented in existing content? Please describe how you would compare change, both temporal and current, against both vector data and map and chart products. Does your system identify modifications, new features, and other discrepancies, generating prioritized alerts for analyst review?

Objective 4: Achieve Seamless Integration with NGA's GEOINT Ecosystem

Is your AFE capability open and modular in nature for integration into NGA operational environment(s)? Describe how you would integrate into the Vector & Cartographic Production System (VCPS) and the Foundation Digital Twin (FDT) data environment for feature registration

and storage. Do you utilize APIs for your AFE capability and what kind of API(s) are they (e.g., REST, SOAP, GraphQL and gRPC)? What standardized formats (e.g., GDB, Shapefile) to support interoperability with other NGA systems and direct dissemination to customer platforms would you recommend? What would your compute and infrastructure needs be for your solution?

Objective 5: Meet and Maintain Stringent Performance and Security Standards

Please describe your solution's scalability, reliability and security standards used in the production of AFE products. Describe how you would achieve processing a standard 1GB source file in less than 10 minutes and achieve 99.9% availability in the NGA production environment. Describe how you would employ cross domain solutions for AFE production.